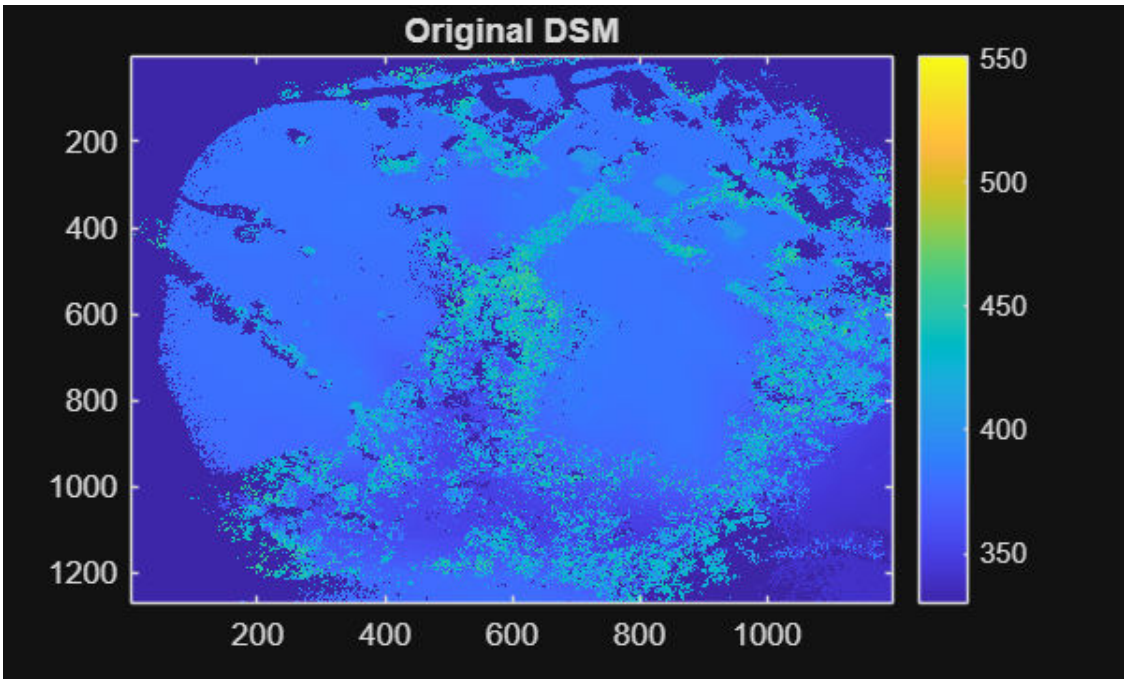
```
clc;
clear;
close all;

%% STEP 1: Load DSM Image (.tif)
dsm = imread('raster.tif');   % Replace with your file name
dsm = double(dsm);          % Convert to double for processing

fprintf('Matrix size: %d x %d\n', size(dsm,1), size(dsm,2));
```

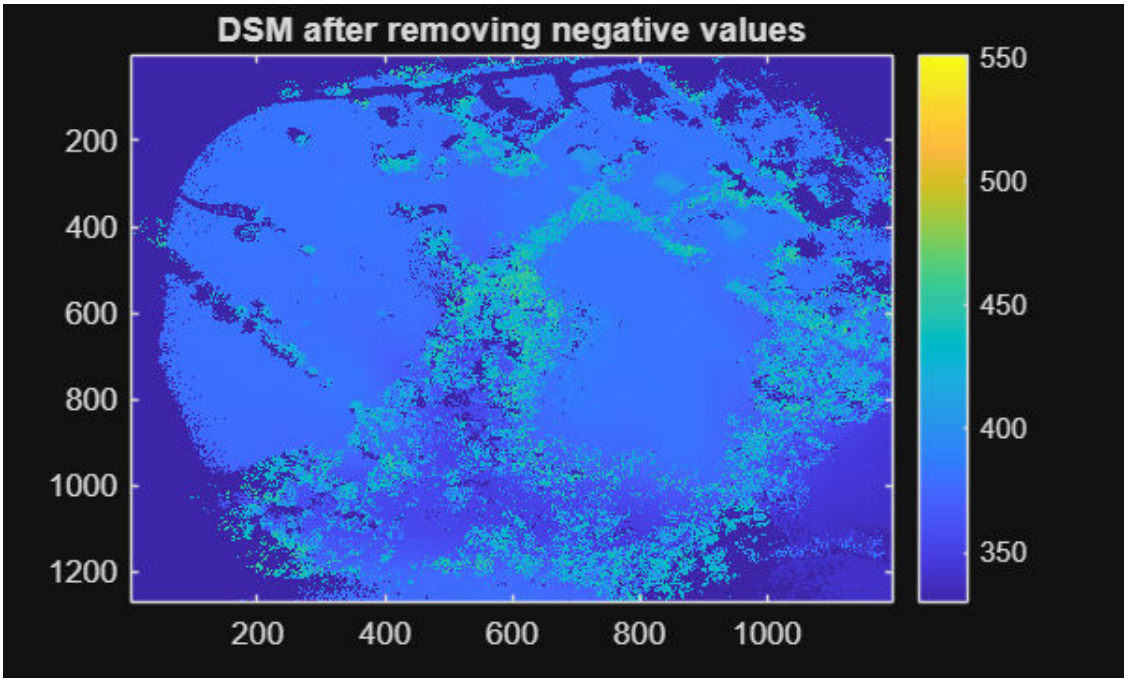
```
Matrix size: 1272 x 1198
```

```
%% STEP 2: Display Original DSM
figure;
imagesc(dsm);
colorbar;
title('Original DSM');
```




```
%% STEP 3: Replace Negative Values with Zero
dsm(dsm < 0) = 0;

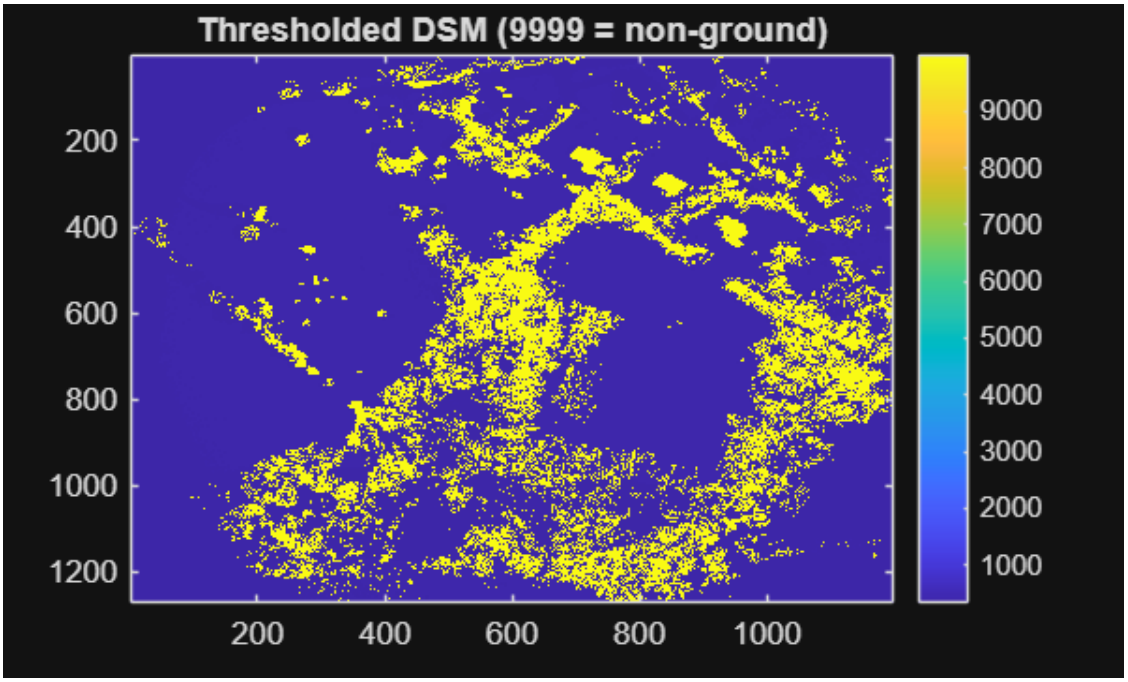
figure;
imagesc(dsm);
colorbar;
title('DSM after removing negative values');
```




```
%% STEP 4: Apply Threshold
threshold = 386;   % <-- Change based on histogram observation

% Set above-ground pixels to 9999
dsm_thresh = dsm;
dsm_thresh(dsm > threshold) = 9999;

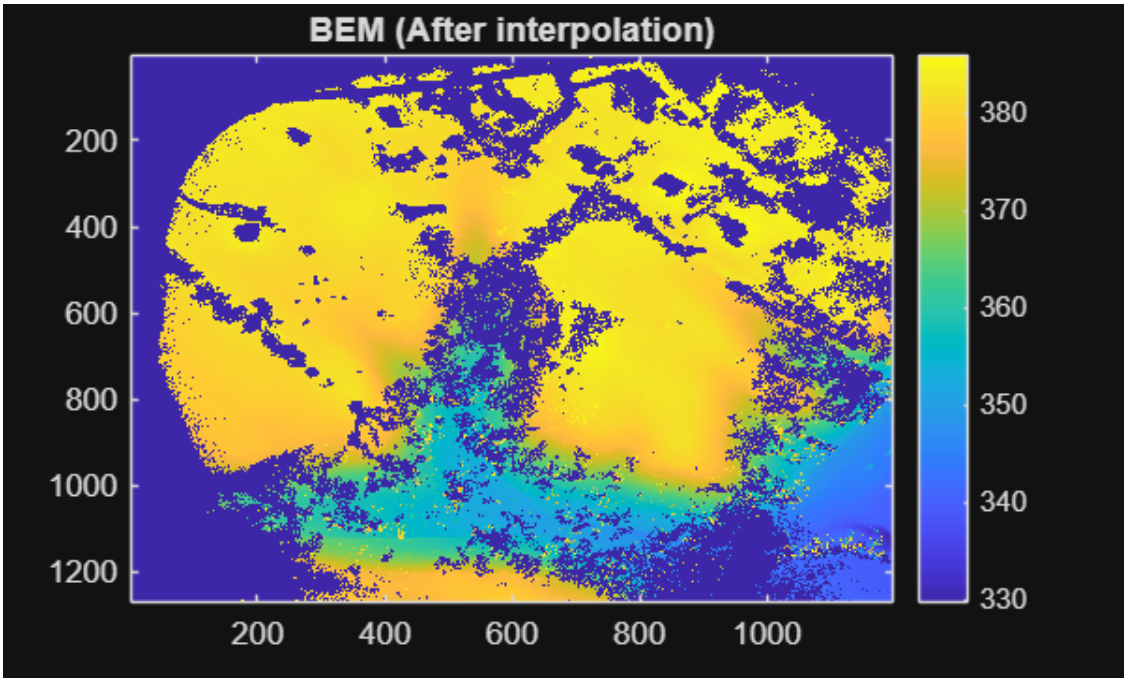
figure;
imagesc(dsm_thresh);
colorbar;
title('Thresholded DSM (9999 = non-ground)');
```




```
%% STEP 5: Create Mask for 9999 Pixels
mask = (dsm_thresh == 9999);

%% STEP 6: Interpolate (Fill 9999 values)
% Using regionfill
bem = regionfill(dsm_thresh, mask);

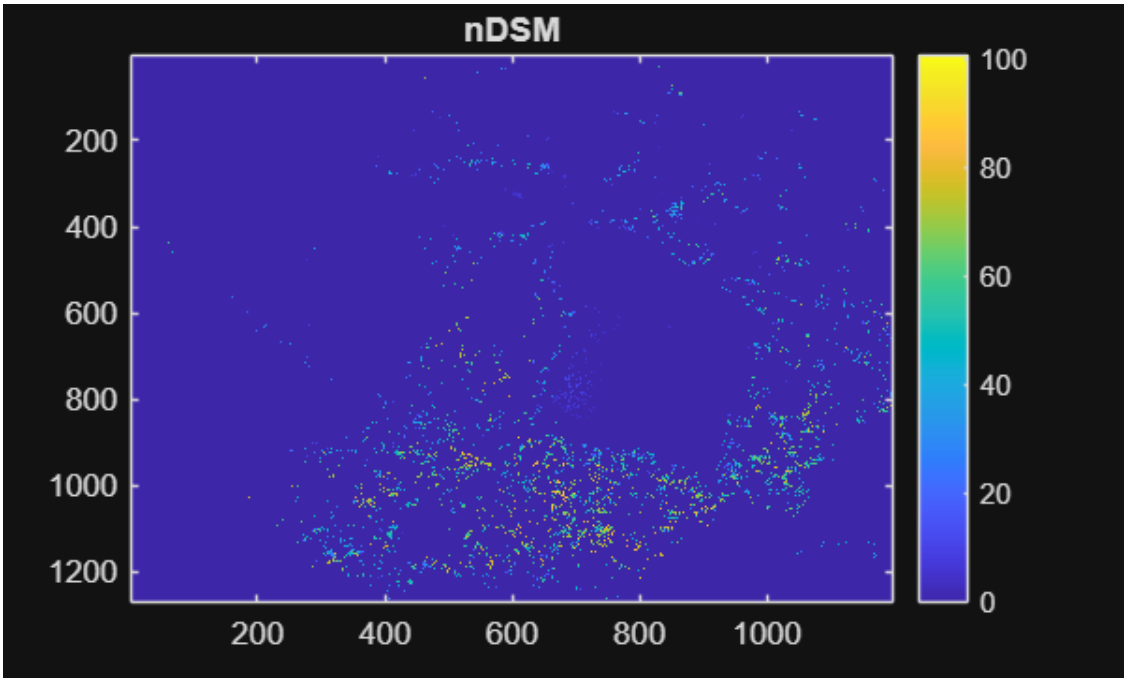
figure;
imagesc(bem);
colorbar;
title('BEM (After interpolation)');
```




```
%% STEP 7: Load Original DSM again (for nDSM calculation)
dsm_original = imread('raster.tif');
dsm_original = double(dsm_original);
dsm_original(dsm_original < 0) = 0;

%% STEP 8: Compute nDSM
ndsm = dsm_original - bem;

figure;
imagesc(ndsm);
colorbar;
title('nDSM');
```




```
%% STEP 9: Display Value Range
fprintf('nDSM Min Value: %.2f\n', min(ndsm(:)));
```

```
nDSM Min Value: 0.00
```

```
fprintf('nDSM Max Value: %.2f\n', max(ndsm(:)));
```

```
nDSM Max Value: 100.87
```

```
%% STEP 10: Save Outputs (Optional)
imwrite(uint16(bem), 'BEM.tif');
imwrite(uint16(ndsm), 'nDSM.tif');
```